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TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Investigation on Precision Sensor Glove for Real-Time Hand Motion Tracking and Visualization

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Guide:

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2024-25

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- The "Precision Sensor Glove for Real-Time Hand Motion Tracking and Visualization" project aims to develop an innovative glove equipped with flex sensors and an IMU for accurate hand motion tracking.
- The glove provides real-time data visualization, enabling precise replication of hand and finger movements in virtual environments.
- This project finds ideal applications in fields of engineering, healthcare and virtual reality, offering a low-cost yet highly effective solution for detailed motion analysis and interaction.

Accurate and real-time tracking of hand movements is crucial for applications in robotics, virtual reality, and remote control systems. Traditional methods often involve expensive and complex equipment, making them inaccessible for widespread use. This project addresses the need for a cost-effective and precise solution by developing a sensor glove that integrates flex sensors and an IMU. The glove enables detailed motion analysis and real-time visualization, providing an affordable and efficient tool for diverse motion tracking applications.

Problem Statement: Accurate and real-time hand movement tracking is essential yet often cost-prohibitive; this project aims to develop an affordable, precise sensor glove using flex sensors and an IMU for diverse applications.

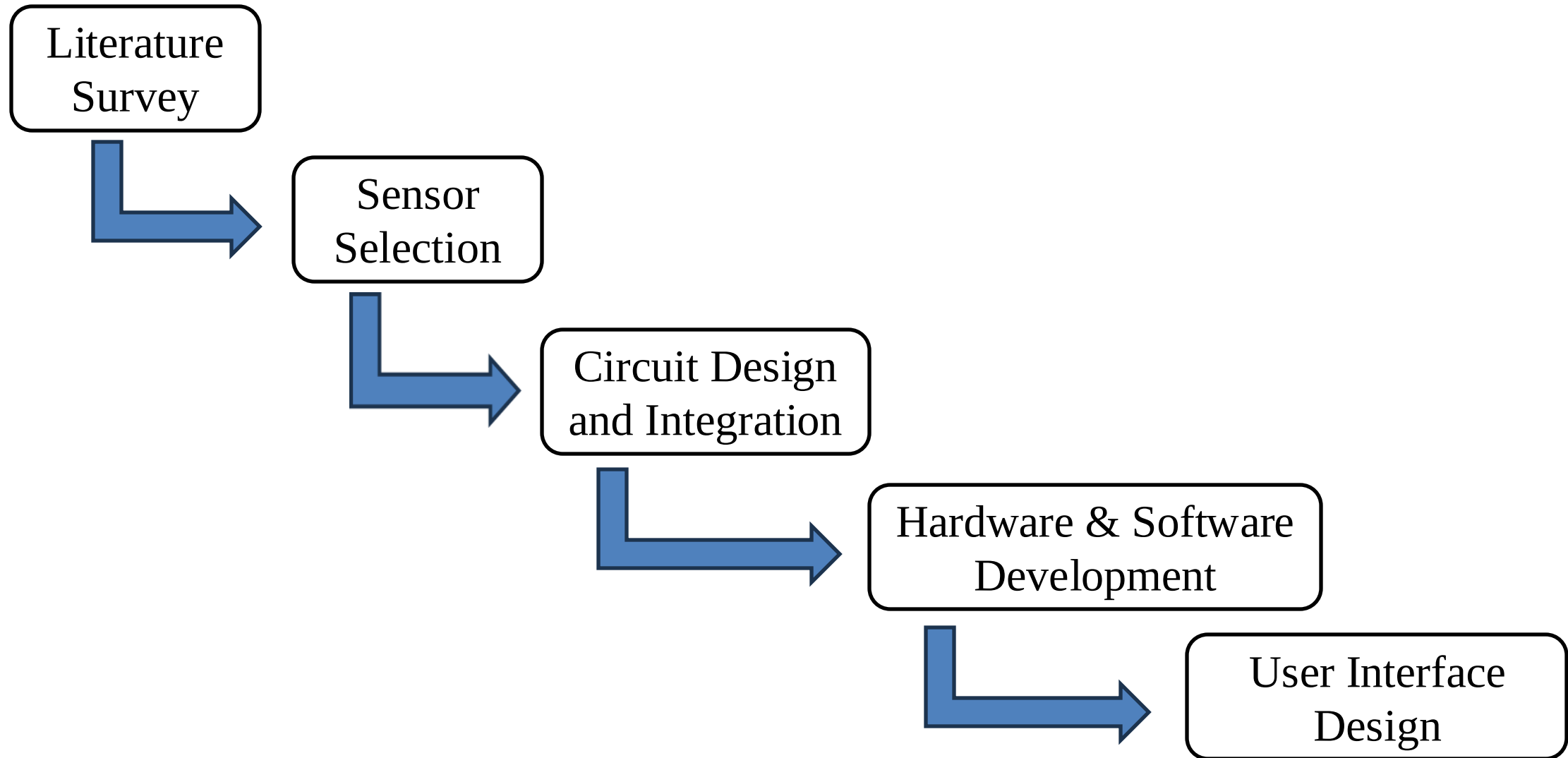
Primary Objectives:

1. To develop a cost-effective sensor glove using flex sensors and an IMU.
2. To enable precise tracking of hand and finger movements in real-time, providing real-time visualization of hand movements

The scope of the project includes:

- To develop a sensor glove capable of accurately tracking and visualizing hand and finger movements in real-time.
- The glove uses flex sensors to detect finger bending and an IMU to measure hand orientation.
- The collected data will be processed and transmitted via ROS for seamless integration with various future applications such as robotics, virtual reality, and remote control systems.
- The project will ensure the glove is user-friendly, cost-effective, and suitable for extended use, with comprehensive documentation to facilitate replication and further development.
- The glove's versatile future applications range from engineering and medical fields, such as surgery training and rehabilitation, to gaming, enhancing user interaction and control in diverse scenarios.

WORK PLAN



ACTION PLAN

Activity Time	July 2024	August 2024	September 2024	October 2024	November 2024
Literature Survey	✓	✓			
Component Sourcing		✓			
Sensor Integration			✓		
Prototype and Software Build-up			✓	✓	
Result Analysis				✓	
Report Writing and Paper Publication					✓

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A smart glove to guide your hands

The wearable device can send tactile feedback to teach users new skills, make robots more dexterous, and help train surgeons and pilots.

By Alex Shipps April 23, 2024

MIT (Massachusetts Institute of Technology) is developing a “smart glove” in order to provide training to surgeons and pilots in the future.



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'Smart glove' can boost hand mobility of stroke patients

January 16, 2024



University of British Columbia is working on a “smart glove” as a future tool for rehabilitation.

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Thank You